Estimands Matter

A Survey-Weighted Reanalysis of Muscle-Strengthening Activity and 10-Year Mortality in NHANES 1999-2006

Comparing population-average and overlap-population estimates

Independent Biostatistics Research
Public NHANES data linked through 2019

Abstract

Objective. To examine how a change in target population affects adjusted associations between baseline muscle-strengthening activity and ten-year mortality. **Methods.** We analyzed 15,091 complete-case adults aged at least 20 from four NHANES cycles. Exposure was at least 8 versus 0-7 self-reported sessions in 30 days. Survey-weighted propensity scores generated population-average (PATE) and overlap-population (PATO) composite weights. Stratified delete-one-PSU jackknife inference re-estimated propensity scores in all 117 replicates. **Results.** There were 1,980 deaths by 120 months. PATE mortality risks were 8.72% versus 8.89%, giving a risk difference of -0.17 (-1.72, 1.38) percentage points and risk ratio 0.98 (0.82, 1.17). PATO risks were 6.41% versus 6.45%, with risk difference -0.04 (-1.06, 0.98) points and risk ratio 0.99 (0.85, 1.17). All intervals are 95% confidence intervals. **Conclusions.** Both adjusted contrasts were near the null but imprecise. Changing the target materially changed participant composition and absolute risks; it did not create compelling evidence of a different relative association. Baseline reports, complete-case selection and unmeasured confounding preclude an established sustained-training causal effect.

Project motivation

Strength activity and mortality in these NHANES cycles have already been studied. Booker and colleagues used the same 2019 linkage but derived activity-specific resistance-training frequency and volume; Boyer and colleagues and Zhao and colleagues provide directly related precedents [1-3]. This project uses that established setting to examine how weighting choices change an analysis.

The motivating question is practical: when the same participants are analyzed with survey-only, population-average and overlap weights, what changes in the target population, balance, effective sample size and ten-year mortality risk? Survey weights describe representation in a survey population. Propensity weights adjust measured differences, and overlap weights also shift the target population [4-6]. Reporting absolute risks and confidence intervals alongside relative contrasts makes these changes easier to interpret.

1 | Design, target and estimation

Component	Analysis specification
Eligibility	Age >=20; eligible linkage; interpretable strength frequency; positive MEC weight; complete covariates. Reported inability excluded.
Time zero / follow-up	MEC examination to 120 months. All surviving analytic participants had at least 154 months of recorded follow-up through 2019.
Exposure	PAD440/PAD460: >=8 versus 0-7 sessions in 30 days. A sessions-based approximation to twice-weekly guidance, not measured adherence to all guideline components.
Outcome / contrast	All-cause death by month 120 inclusive. Marginal risk difference and risk ratio; conditional Cox HR as benchmark.
Interpretation	Adjusted baseline association within a restricted response domain. An ideal sustained-strategy trial cannot be emulated with one baseline report.

Covariates and missingness

The propensity model included age, sex, race/ethnicity, education, income/poverty ratio, smoking, alcohol, BMI, broad aerobic activity, self-rated health, diabetes, hypertension, major cardiovascular disease, cancer and cycle. Natural splines used 4 df for age and 3 each for BMI and poverty ratio. The realized knots are held fixed for reproduction and replicate fitting. Other variables were categorical. Alcohol distinguished lifetime <12 drinks, no past-year drinking and past-year drinking; aerobic activity distinguished neither, moderate only, vigorous only and both. These coarse measures leave residual confounding.

The working DAG recognizes that baseline disease, health and BMI can affect recent exercise and mortality, while also reflecting earlier activity. Their adjustment therefore supports a baseline association, not a total effect of lifetime exercise. Complete-case analysis used no missing categories or univariate significance screening; a missingness audit describes the resulting selection. The PATE target is explicitly the survey-weighted complete-response domain, which may differ from adults excluded for missing information.

Weights and uncertainty

MEC8YR equaled WTMEC4YR/2 for 1999-2002 and WTMEC2YR/4 for 2003-2006. Original masked strata and nested PSUs were retained: 58 strata, 117 PSUs and 59 design degrees of freedom. Let d be MEC8YR and e the survey-logistic propensity score. PATE composite weights were d/e for exposed and d/(1-e) for unexposed participants. PATO weights were d(1-e) and de. Separate-arm weighted means estimated risks; PATO targets a covariate distribution tilted by e(1-e). [4-7]

Primary inference used stratified JKn replicates generated before domain restriction, refitting PS coefficients in each replicate with the fixed basis. Intervals used t(59), logit risks, direct risk differences and log risk ratios. Conventional survey-logistic standardization refit both its outcome model and averaging distribution in replicates. Fixed-PS Taylor estimates were a comparator. These methods propagate sampling and PS coefficient uncertainty, not model-choice, nonresponse or linkage uncertainty. [6-7]

The DAG, covariate dictionary, missing-data plan, diagnostics and sensitivity rules describe the methods; DESIGN_FREEZE.md records the specification and file hashes. These files document the implemented analysis, not a prospective public registration. Target-trial components guide the question, but the sustained-exposure gap remains. [8]

2 | Analytic sample and selection

Selection step	Retained n	Excluded at step
All NHANES participants	41,474	--
Adults age 20+	20,311	21,163
Mortality linkage eligible	20,290	21
Interpretable strength exposure	19,420	870
Positive MEC weight	18,264	1,156
Covariate-complete analytic sample	15,091	3,173

Of 870 exposure exclusions after linkage, 851 reported inability to perform strength activity; the other 19 had unavailable or uninterpretable responses. All complete cases had valid 120-month outcomes, so no further outcome exclusions occurred. Excluding inability restricts the population and should not be mistaken for a nationally unrestricted contrast.

Table 1. Survey-weighted baseline characteristics

Characteristic	All	<8 sessions	>=8 sessions
Unweighted n	15,091	12,265	2,826
Deaths by 120 months, n	1,980	1,732	248
Age, mean years	45.7	46.6	42.6
Female, %	51.2	52.8	45.3
Non-Hispanic White, %	73.4	72.7	75.8
College graduate, %	25.9	22.9	36.9
Income/poverty ratio, mean	3.1	3.0	3.5
BMI, mean kg/m 2	28.2	28.6	26.9
Current smoking, %	24.4	26.0	18.3
Past-year drinking, %	71.5	69.0	80.8
Both aerobic intensities, %	25.0	17.9	51.1
Poor self-rated health, %	2.7	3.2	1.0
Diagnosed diabetes, %	6.5	7.1	4.4
Major CVD, %	8.0	8.6	5.6
Cancer history, %	7.9	8.2	7.1

Weighted exposure prevalence was 21.37%. The complete-case MEC weight sum was 167.5 million, a response-domain total, not a newly calibrated estimate of all adults. Full category-level denominators appear in the supplemental CSV tables.

Complete cases retained 82.63% of the 18,264 otherwise eligible examined adults. Missingness was largest for alcohol (8.01%), poverty ratio (7.63%) and BMI(2.58%). Excluded respondents had lower observed mean poverty ratio (2.57 versus 3.07), more poor health (3.75% versus 2.69%) and lower exposure prevalence (18.77% versus 21.37%). Original survey weights do not resolve this item-nonresponse selection.

3 | Balance, overlap and target population

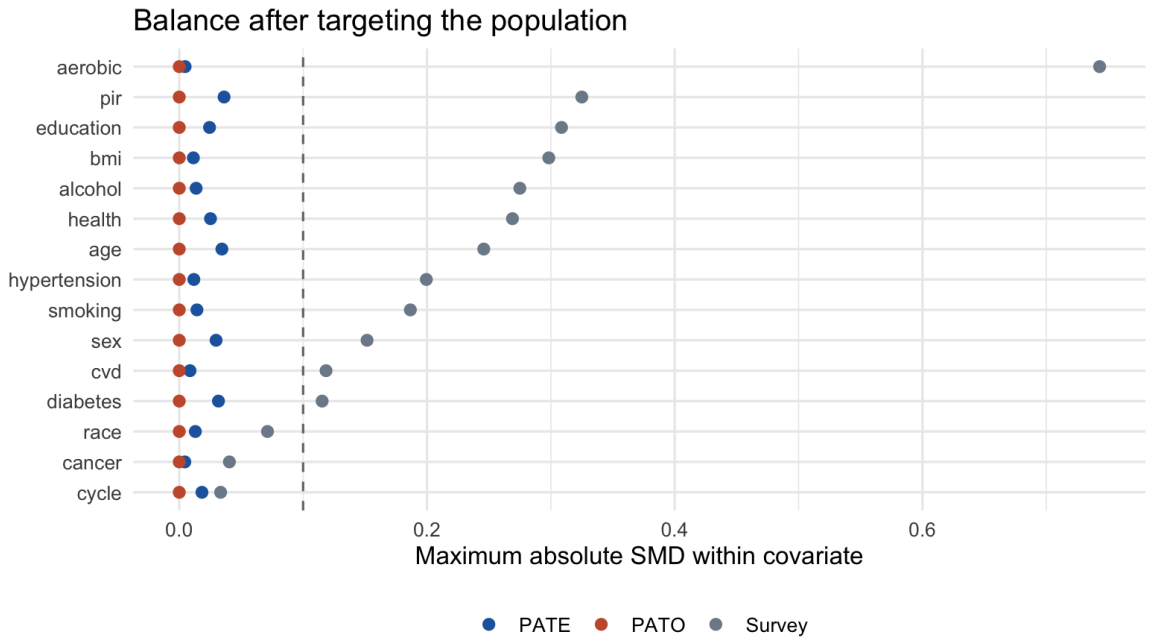


Figure 1. Maximum absolute SMD within each covariate (including spline/dummy columns). The full basis-level diagnostic is retained separately; .10 is the descriptive threshold used here.

The largest absolute SMD fell from 0.743 under survey weights alone to 0.036 under PATE weights and approximately zero under PATO weights. The latter is expected for included model columns from the weighted logistic score equations; exact mean balance is not evidence that unmeasured confounding is absent. The unweighted-PS comparator had maxima 0.051 and 0.052 and was not selected.

Weighting	ESS: <8	ESS: >=8	Max within-arm weight / mean
Survey	7,774	2,013	4.8
PATE	7,203	1,194	20.9
PATO	4,442	1,957	14.0

Observed propensity scores ranged from 0.014 to 0.717. The empirical common range was 0.022-0.702; 69 participants (0.33% of survey weight) lay outside it. No hard support trimming was applied. The maximum exposed PATE multiplier was 45.42. Bounded overlap multipliers avoid inverse-propensity explosions, but composite weights still inherit survey-weight variability. ESS describes dispersion, not independent sample size or guaranteed precision.

Who receives more representation under PATO?

Characteristic	Original domain	PATE composite	PATO composite
Mean age	45.7	45.9	43.7
College graduate, %	25.9	25.9	33.5
Both aerobic intensities, %	25.0	25.2	41.8
Poor health, %	2.7	2.6	1.2

Composite columns pool exposure arms after separate normalization. PATO represents a younger, more educated and more aerobically active population with less poor health. The additional e(1-e)-tilted population table gives a closely corresponding model-based target. Neither PATO nor the complete-case PATE is automatically representative of all US adults.

4 | Ten-year marginal mortality risks

Analysis	Risk >=8, %	Risk <8, %	RD, points (95% CI)	RR (95% CI)
Survey only	5.55	9.79	-4.24 (-5.39, -3.09)	0.57 (0.47, 0.69)
Logistic standardization	9.11	8.85	0.26 (-1.12, 1.64)	1.03 (0.89, 1.20)
PATE	8.72	8.89	-0.17 (-1.72, 1.38)	0.98 (0.82, 1.17)
PATO	6.41	6.45	-0.04 (-1.06, 0.98)	0.99 (0.85, 1.17)

All analyses use the same 15,091 participants and 1,980 events. Estimates compare >=8 with <8 sessions/30 days. Risks have corresponding 95% CIs in the complete results table. Standardization is a conventional outcome-model reference, not a doubly robust estimator.

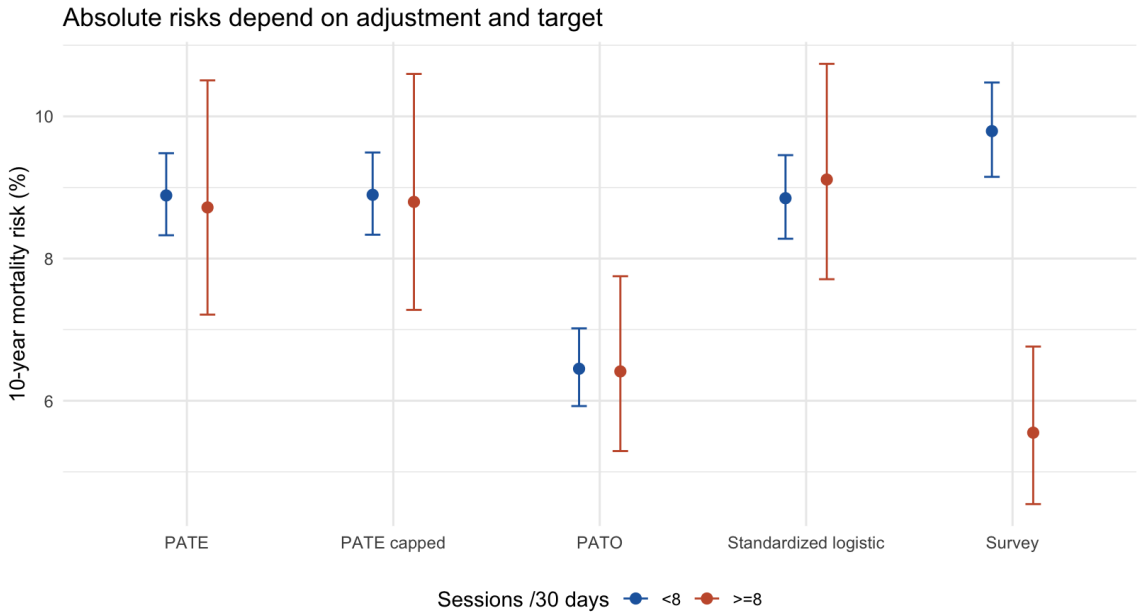


Figure 2. Absolute risks with jackknife confidence intervals. The capped PATE result is a stability check.

Interpretation of the contrasts

The unadjusted survey contrast was -4.24 (-5.39, -3.09) percentage points, substantially more negative than either propensity-adjusted contrast. This attenuation is consistent with the large measured baseline differences; it does not quantify the amount of confounding removed. PATE gave -0.17 (-1.72, 1.38) points; its interval permits a modest lower or higher risk. PATO gave -0.04 (-1.06, 0.98) points, also compatible with either direction. Non-significance does not establish equivalence or absence of an effect.

PATO arm risks were around 6.45%, compared with roughly 8.89% for PATE. This illustrates a change of target distribution, not a mortality reduction produced by the choice of weighting algorithm. The PATE and PATO estimates are correlated analyses of the same observations. We did not infer effect heterogeneity from their numerical difference or from comparing significance labels.

Uncertainty cross-check

Direct weighted ratios and survey means agreed to numerical precision. Jackknife RD standard errors were 0.773 and 0.511 percentage points for PATE and PATO; fixed-PS Taylor values were 0.962 and 0.669. Propensity estimation can increase or decrease variance, so treating estimated weights as fixed is not uniformly conservative or anti-conservative. Refitting accounts for propensity coefficient estimation; narrower intervals alone do not justify a variance method.

5 | Sensitivity analyses and Cox benchmark

Sensitivity / target	n / events	RD, points (95% CI)	RR (95% CI)
24-month survivor / PATE	14,794 / 1,683	-0.19 (-1.61, 1.23)	0.98 (0.81, 1.18)
24-month survivor / PATO	14,794 / 1,683	-0.05 (-0.97, 0.88)	0.99 (0.84, 1.17)
No cancer/CVD / PATE	12,581 / 1,009	-0.53 (-1.66, 0.60)	0.90 (0.71, 1.14)
No cancer/CVD / PATO	12,581 / 1,009	0.03 (-0.86, 0.92)	1.01 (0.80, 1.27)
99th-percentile cap / PATE	15,091 / 1,980	-0.10 (-1.65, 1.45)	0.99 (0.83, 1.18)

The survivor analysis removes 297 early deaths and estimates risk during months 24-120 among those surviving 24 months: an eight-year landmark endpoint, not the original unconditional ten-year risk. Disease exclusion changes the target again. Both PS models were refit using the same functional forms; balance remained within .10. These restrictions cannot rule out reverse causation and may introduce selection.

The weight-stability analysis uses a single cap on PATE propensity multipliers. Capping them at their within-arm 99th percentile changed PATE RD from -0.17 to -0.10 points; exposed ESS increased to 1,266. The observed estimate was not materially dependent on this cap, but the capped estimator no longer exactly targets PATE. No alternative cutoffs were searched.

Hazard ratio and proportional hazards

The adjusted survey Cox HR was 0.93 (95% CI 0.77, 1.13). An exposure-by-follow-up-period test gave p=0.442, with HRs 1.02 in months 0-60 and 0.88 in months 61-120. This did not detect that particular departure from a constant exposure HR. It cannot establish proportional hazards for the entire model.

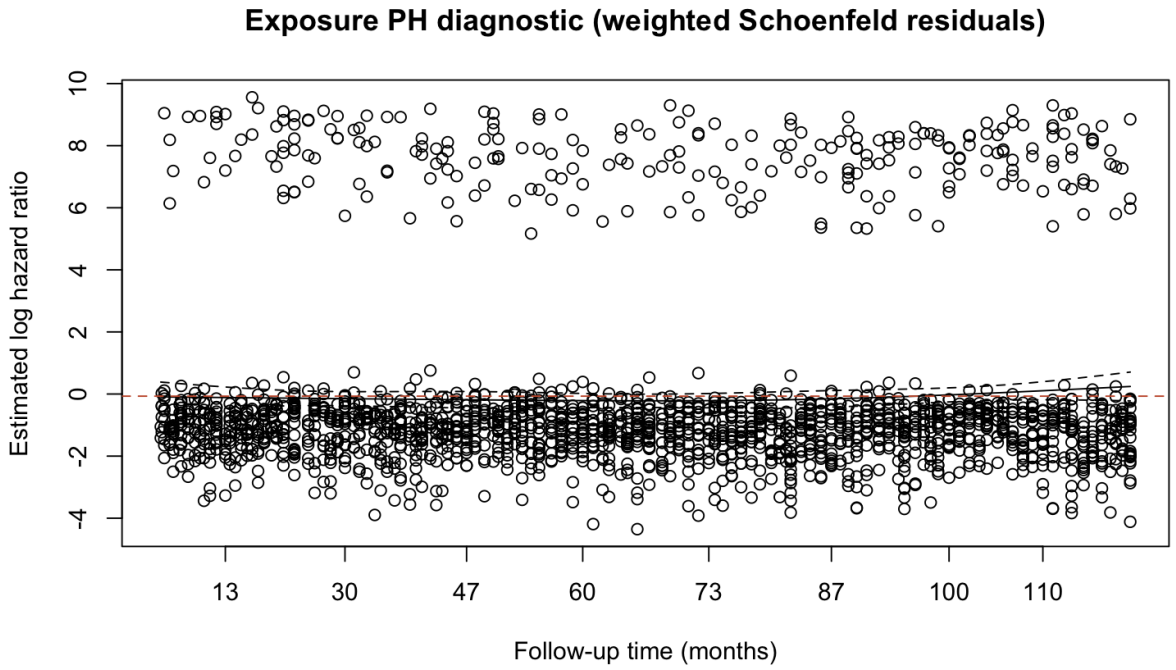


Figure 3. Weighted exposure Schoenfeld diagnostic. This plot is exploratory; its ordinary residual-test p-values do not fully incorporate the complex survey design.

The exploratory global Schoenfeld test was p=0.00019, suggesting non-proportionality somewhere in the adjusted model. Consequently the single benchmark HR requires caution and should not be read as a constant ten-year risk ratio. The fixed-horizon marginal risk analyses remain primary. Two month 0 deaths were assigned 0.5 months only for Cox bookkeeping; their binary endpoints were unchanged.

6 | What the findings support

A target-population result, not a new exercise verdict

The main methodological finding is that good balance and a change of target can coexist with similar relative contrasts. PATO improved the exposed group's weight-based ESS and described a population with lower absolute mortality risk, while the unexposed group's ESS decreased. There is no general rule that overlap weighting increases every group's ESS. The reason to use it is alignment with a scientifically defensible overlap population, not an expectation of a favorable p-value.

PATE and PATO here are conventional causal-estimand labels, but the reported numbers are assumption-dependent adjusted associations. A causal interpretation would require adequate control of common causes, positivity in the relevant target, consistent and well-defined exposure versions, and valid survey/response selection assumptions. Strong balance on measured covariates alone does not meet these conditions.

Material limitations

Exposure and temporal order. PAD460 records sessions, not necessarily different days. Training load, sets, duration and sustained adherence are unobserved. PAD440 can include push-ups or sit-ups, and the eight-session rule is only an operational approximation. Interview measurements and MEC examination are not perfectly simultaneous. We cannot identify a long-term assigned strength-training strategy from a one-time recent-activity report.

Confounding and reverse causation. Latent frailty, disability short of reported inability, disease severity, diet, healthcare access and health consciousness may influence activity and mortality. Broad aerobic and alcohol measures leave dose confounding. Baseline BMI and disease may be consequences of earlier exercise as well as predictors of current behaviour. The chosen adjustment does not identify the total effect of exercise across the life course. Early-death and disease-exclusion analyses address only limited aspects of these problems.

Selection and generalizability. Restriction to complete cases removed 17.37% of otherwise eligible examined participants. Income and health profiles differed between retained and excluded respondents. Original MEC weights do not repair that loss. Excluding reported inability further narrows the target. PATO changes that target again. These historical 1999-2006 response domains should not be presented as the present-day US population.

Outcome measurement and inference. Public mortality files preserve final vital status but replace selected follow-up times with synthetic values. This can affect classification near 120 months and survival diagnostics. Linkage eligibility and possible misclassification remain relevant. JKn uses masked strata/PSUs and a with-replacement design approximation; it conditions on selected model forms and cannot propagate unknown linkage or nonresponse mechanisms. The Cox PH evidence is mixed and partly exploratory.

Project scope and possible extensions

The comparison focuses on how survey and propensity weighting affect population composition, balance, effective sample size, ten-year absolute risks and interval estimates. It provides a reproducible worked example of those calculations. Future extensions could evaluate multiple imputation or response weighting with explicit assumptions; these methods are not included here. Better longitudinal exposure data would be needed to study sustained training strategies or support stronger causal claims.

No conclusion here constitutes individual medical or training advice. The absence of a clear adjusted association in this analysis is not evidence that strength training lacks established benefits or that every plausible mortality effect is zero.

7 | Sources, reproducibility and responsibility

1. Booker R et al. Associations Between Resistance Training and All-Cause Mortality: NHANES 1999-2006. Published online 2024. Source.

2. Boyer W et al. Independent and combined effects of aerobic physical activity and muscular strengthening activity on all-cause mortality. J Phys Act Health. 2020; 17: 881-888. Source.

3. Zhao G et al. Leisure-time aerobic physical activity, muscle-strengthening activity and mortality risks among US adults: NHANES linked mortality study. Br J Sports Med. 2014. Source.

4. Li F, Morgan KL, Zaslavsky AM. Balancing Covariates via Propensity Score Weighting. JASA. 2018; 113: 390-400. Source.

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6. Zeng Y, Li F, Tong G. Moving toward best practice when using propensity score weighting in survey observational studies. arXiv preprint 2501.16156, revised 2026. Source.

7. Lumley T. Analysis of Complex Survey Samples. J Stat Softw. 2004; 9(8); survey package 4.5 and JKn documentation. Source.

8. Hernan MA, Robins JM. Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available. Am J Epidemiol. 2016; 183: 758-764. Source.

9. NCHS. NHANES weighting and variance tutorials; cycle-specific official codebooks. Source.

10. NCHS.2019 Public-Use Linked Mortality Files data dictionary and linkage methodology, 2022. Source.

11. Zubizarreta JR. Stable Weights that Balance Covariates for Estimation With Incomplete Outcome Data. JASA. 2015; 110: 910-922. Design/balance motivation; method not implemented. Source.

Reproduce and inspect

Run python 3 run_project.py from the project root; add --download to fetch or validate official originals. Source-file SHA-256 manifests, exact variable mappings, analysis specifications, references, code, complete tables and plots are included. R 4.5.2 and survey 4.5 were used. The tests independently check outcome parsing, weighted-risk arithmetic, PS-fit agreement, replicate covariance and Cox coefficients. Published result tables are checked against the reference hashes in docs/DESIGN_FROZEN.json.

Core artifacts: DESIGN_FREEZE.md; docs/PROTOCOL.md; MISSING_DATA_PLAN.md; VARIABLE_DICTIONARY.md; data/download_manifest.csv; outputs/tables/main_results.csv; outputs/tables/sensitivity_results.csv; outputs/tables/validation_results.csv. Supplementary diagnostics include complete-case comparisons, cycle weights, full SMDs, support and weight quantiles.

Data and interpretation

This independent project uses public NHANES data. References identify the data and methods, and the reported estimates are adjusted baseline associations subject to the limitations described above.